

```
def quick_sort(arr):
    if len(arr) <= 1:
        return arr

    pivot = arr[0]

    left = []
    right = []

    for x in arr[1:]:
        if x <= pivot:
            left.append(x)
        else:
            right.append(x)

    return quick_sort(left) + [pivot] + quick_sort(right)

arr = list(map(int, input("Enter elements separated by space: ").split()))

print("Before sorting:", arr)

arr = quick_sort(arr)

print("After sorting:", arr)

print("Best Case      = O(n log n)")
print("Average Case = O(n log n)")
print("Worst Case     = O(n^2)")
```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:59)
on win32
Enter "help" below or click "Help" above for more information

>>>

==== RESTART: C:/Users/ROOPA SAI DIVYA/OneDrive/Documents
Enter elements separated by space: 8 3 5 1 9 2
Before sorting: [8, 3, 5, 1, 9, 2]
After sorting: [1, 2, 3, 5, 8, 9]
Best Case = O(n log n)
Average Case = O(n log n)
Worst Case = O(n^2)

>>>

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